

An Algorithm for the Distance Between Two Finite Sequences

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1. THE ALGORITHM

In a biological problem, which will be described later, it is necessary to compute the distance or degree of dissimilarity between two finite sequences. A mathematical definition of this distance was brought to my attention by S. M. Ulam, and an algorithm for computing it will be presented here. If m and n are the lengths of the two sequences and $m \leq n$, then the number of computational steps in the algorithm is m^2n , where each step consists of selecting the largest of three known numbers. In Section 2 it will be shown how the algorithm can be changed to compute the modifications of this distance which are required in the biological context.

DEFINITION. The distance between $(a_1, a_2, \dots, a_m)$ and $(b_1, b_2, \dots, b_n)$ is the smallest number of steps necessary to make the sequences identical, where a step is

- (i) the deletion of a term from either sequence, or
- (ii) the change of a term in either sequence, so as to make it equal to a term in the other sequence.

As an example of this definition, the distance between the sequences (a, b, c, a) and (a, c, b, b, a) is 2, because it takes at least 2 steps to make them alike:

1. Change c to b in the first sequence.
2. Delete c from the second sequence.

If we had tried to make these sequences alike by deletions only, it would have taken 3—say, the deletion of each b .

The following theorem gives an algorithm for computing the distance in general, and it is based on a procedure which was introduced by D. Sankoff (1):

THEOREM 1. *The distance between $(a_1, a_2, \dots, a_m)$ and $(b_1, b_2, \dots, b_n)$, when $m \leq n$, is*

$$\min\{k + m + n - 2v_k(a_m, b_n) \mid k = 0, 1, \dots, m\},$$

where $v_k(a_m, b_n)$ is defined inductively as follows:

(i) $v_0(a_1, b_1) = 1$ or 0 according as a_1 equals a term of $(b_1, b_2, \dots, b_j)$ or not.

(ii) $v_0(a_i, b_1) = 1$ or 0 according as b_1 equals a term of $(a_1, a_2, \dots, a_i)$ or not.

(iii) $v_k(a_1, b_j) = 1 = v_k(a_i, b_1)$ for $k > 0$.

(iv) $v_k(a_i, b_j) = \max\{v_k(a_{i-1}, b_j), v_k(a_i, b_{j-1}), 1 + v_{k-t}(a_{i-1}, b_{j-1})\}$ where $t = 0$, if $a_i = b_j$, and $t = 1$, if $a_i \neq b_j$.

(v) $v_{-1} \equiv -1$.

Each of $v_0, v_1, v_2, \dots$, constructed in this theorem, is itself a distance function defined on pairs of sequences. A whole variety of distance functions, including the one in the theorem, can be expressed as functions of $v_0, v_1, v_2, \dots$. Therefore, that part of the proof of Theorem 1 which gives the combinatorial interpretation of the number $v_k(a_i, b_j)$ is sufficiently important to state as a separate lemma:

LEMMA. *Let $p_k(a_i, b_j)$ be an order-preserving one-to-one correspondence between a subsequence of $(a_1, a_2, \dots, a_i)$ and a subsequence of $(b_1, b_2, \dots, b_j)$, where $p_k(a_i, b_j)$ contains as many pairs as possible, subject only to the constraint that there be at most k pairs with unequal parts; the number of pairs in such a correspondence is $v_k(a_i, b_j)$, as defined in Theorem 1.*

Proof. It must be shown that, if $v_k(a_i, b_j)$ is defined as the number of pairs in a correspondence $p_k(a_i, b_j)$, then it satisfies conditions (i) through (iv) in the theorem. The first three conditions are obvious. For instance, condition (i) says that for the sequences (a_1) and $(b_1, b_2, \dots, b_j)$ the correspondence $p_0(a_1, b_j)$ has at most one pair (a_1, b_g) in it, and then only if $a_1 = b_g$; there cannot be an unequal pair, because $k = 0$.

The proof will be concluded by proving condition (iv), which says that

$$v_k(a_i, b_j) = \max\{p, q\},$$

where

$$p = 1 + v_{k-t}(a_{i-1}, b_{j-1})$$

and

$$q = \max\{v_k(a_{i-1}, b_j), v_k(a_i, b_{j-1})\}.$$

This can be reduced to three cases: First, if

$$v_k(a_i, b_j) = p = q,$$

then there is nothing to prove. Second, if

$$v_k(a_i, b_j) > q,$$

then both a_i and b_j belong to pairs in any $p_k(a_i, b_j)$. Since the correspondence is order-preserving, a_i and b_j must be paired to each other. The remaining pairs belong to $p_{k-t}(a_{i-1}, b_{j-1})$. Therefore,

$$v_k(a_i, b_j) = p.$$

Third, if

$$v_k(a_i, b_j) > p,$$

then $p_k(a_i, b_j)$ cannot be constructed from $p_{k-t}(a_{i-1}, b_{j-1})$ by adding on the pair (a_i, b_j) . Therefore, either a_i or b_j fails to appear in any pair in $p_k(a_i, b_j)$. Therefore,

$$v_k(a_i, b_j) = q.$$

This proves the lemma.

Proof of Theorem 1. Let $p_k(a_m, b_n)$ be a correspondence between subsequences of $(a_1, a_2, \dots, a_m)$ and $(b_1, b_2, \dots, b_n)$, as defined above. The number of terms not involved in this correspondence is given by

$$m + n - 2v_k(a_m, b_n).$$

If these terms are deleted from the two sequences, we are left with two subsequences which are in a one-to-one correspondence with at most k unequal pairs. By changing at most k pairs the subsequences are made equal. The total number T of changes and deletions in this process reduces to two cases:

- (i) If $p_k(a_m, b_n)$ has exactly k unequal pairs, then

$$T = k + m + n - 2v_k(a_m, b_n).$$

(ii) If $p_k(a_m, b_n)$ has k' unequal pairs, where $k' < k$, then

$$T = k' + m + n - 2v_k(a_m, b_n).$$

For each k we have already determined $v_k(a_m, b_n)$ in such a way that T will be minimal, and, therefore, it remains to find that k for which T is smallest. It is obvious that such a value of k is among those which satisfy case (i), and, therefore, we see that

$$\min\{k + m + n - 2v_k(a_m, b_n) \mid k = 0, 1, \dots, m\}$$

is the smallest possible value of T . This proves the theorem.

2. AN APPLICATION IN BIOLOGY

A method of computing the distance between two sequences is needed in biology for the determination of the *evolutionary distance* between two homologous proteins. Each protein is characterized by an amino acid sequence, and the distance between two of them can be defined as the minimum number of mutations which would account for the conversion of one sequence into the other, or, if they are not of the same length, the minimum number of mutations which would account for the conversion of the shorter one into a subsequence of the longer one. This is a simpler definition of distance than the one calculated in Theorem 1, because it assumes that the deletions are so much less likely to occur in evolution than mutations that the number of deletions is held at its minimum. That is, if one sequence is longer than the other, then the number of deletions equals their length difference, and these deletions are distributed over the longer amino acid sequence, so as to minimize the number of mutations making the remaining sequences identical. This distance is computed as follows:

Let $(a_1, a_2, \dots, a_m)$ and $(b_1, b_2, \dots, b_n)$ be amino acid sequences; the distance between them is the smallest value of k such that

$$v_k(a_m, b_n) = \min\{m, n\},$$

where v_k is defined inductively as in Theorem 1 except that the letter t is now understood as the minimum number of mutations in genetic material which would account for the conversion of amino acid a_i into amino acid b_j . This number is always 0, 1, 2, or 3, and it has been tabulated for all amino acid pairs by W. M. Fitch and E. Margoliash (2). With this modification in the inductive step of the definition of v_k it becomes necessary to redefine the initial cases v_0 , v_1 , and v_2 , and it is obvious how

to do so, if we recognize that $v_k(a_i, b_j)$ is now the length of the longest common subsequence of $(a_1, a_2, \dots, a_i)$ and $(b_1, b_2, \dots, b_j)$ after both sequences are converted in any way which can be accounted for by k or less mutations in their genetic code.

The computations of distance, which have been shown here, are constructive, in that each $v_k(a_m, b_n)$ has been found by a method which displays every one-to-one correspondence $p_k(a_m, b_n)$ between subsequences of the two given sequences. Therefore, in the biological context we have arrived at a display of all the amino acid sequences which are most likely to have been the *common ancestor* of the two given ones.

Note. Albert Nijenhuis and David Sankoff both observed that the algorithm of this paper can be modified so as to calculate in mn , rather than mn^2 steps, the distance

$$d(a_1 a_2 \cdots a_m, b_1 b_2 \cdots b_n)$$

between two sequences. This distance is defined as the minimum total number of term deletions and term changes in the sequences $a_1 a_2 \cdots a_m$ and $b_1 b_2 \cdots b_n$ required to make them identical.

It is evident that $d(a_i, b_j)$ is 0 or 1 according as a_i and b_j are alike or not. Furthermore, if we introduce the empty sequence ϕ , then for $i = 1, \dots, m$ and $j = 1, \dots, n$ we can say:

$$\begin{aligned} d(\phi, \phi) &= 0, \\ d(a_1 \cdots a_i, \phi) &= i, \\ d(\phi, b_1 \cdots b_j) &= j. \end{aligned}$$

Using these as starting values, we can compute $d(a_1 \cdots a_m, b_1 \cdots b_n)$ by $m \cdot n$ applications of the following recursion:

$$\begin{aligned} d(a_1 \cdots a_i, b_1 \cdots b_j) \\ = \min\{d(a_1 \cdots a_{i-1}, b_1 \cdots b_j) + d(a_i, \phi), d(a_1 \cdots a_i, b_1 \cdots b_{j-1}) \\ + d(\phi, b_j), d(a_1 \cdots a_{i-1}, b_1 \cdots b_{j-1}) + d(a_i, b_j)\}. \end{aligned}$$

This distance is a metric, and it will continue to be so, if, instead of making $d(x, y)$ equal to 0 or 1, we let it be the distance in any metric space containing the points $\{\phi, a_1, \dots, a_m, b_1, \dots, b_n\}$. Then the initial values are

$$d(a_1 \cdots a_i, \phi) = \sum_{h=1}^i d(a_h, \phi)$$

and

$$d(\phi, b_1 \cdots b_j) = \sum_{h=1}^j d(\phi, b_h),$$

while the recurrence relation is exactly as before.

REFERENCES

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